

Stochastic Orders, Strassen's Theorem, and Optimal Transport

1 Overview

A stochastic order is an order relation between probability measures. Instead of comparing two points x and y , one compares two laws μ and ν by asking whether ν is larger, more variable, or more spread out than μ , according to some chosen class of test functions.

Strassen's theorem is the bridge between two viewpoints:

- an *analytic* viewpoint, where one compares integrals of test functions against μ and ν ;
- a *probabilistic* viewpoint, where one asks whether there are random variables $X \sim \mu$ and $Y \sim \nu$ on the same probability space satisfying a pathwise constraint, such as $X \leq Y$ almost surely.

Optimal transport enters because a joint law of (X, Y) with marginals μ and ν is exactly a transport plan between μ and ν . Thus Strassen's theorem can be read as a feasibility theorem for constrained transport plans.

2 Couplings and transport plans

Let E be a measurable space, and let $\mu, \nu \in \mathcal{P}(E)$ be probability measures. A *coupling* of μ and ν is a probability measure $\pi \in \mathcal{P}(E \times E)$ such that

$$\pi(A \times E) = \mu(A), \quad \pi(E \times A) = \nu(A)$$

for every measurable set $A \subseteq E$. The set of all couplings is denoted by

$$\Pi(\mu, \nu).$$

Equivalently, $\pi \in \Pi(\mu, \nu)$ is the joint law of a pair of random variables (X, Y) such that

$$X \sim \mu, \quad Y \sim \nu.$$

In optimal transport, the same object is called a *transport plan*. Given a cost function $c : E \times E \rightarrow (-\infty, \infty]$, the Kantorovich optimal transport problem is

$$\mathcal{T}_c(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \int_{E \times E} c(x, y) \pi(\mathrm{d}x, \mathrm{d}y).$$

Strassen's theorem concerns the existence of couplings satisfying structural constraints. This is a transport problem with constraints rather than only a cost minimisation problem.

3 Stochastic orders generated by test functions

Definition 1 (Order generated by test functions). *Let $\mathcal{F}$ be a class of measurable real-valued functions on E . We write*

$$\mu \preceq_{\mathcal{F}} \nu$$

if

$$\int_E f \, d\mu \leq \int_E f \, d\nu \quad \text{for every } f \in \mathcal{F}$$

whenever the integrals are well-defined.

Different choices of $\mathcal{F}$ give different stochastic orders.

3.1 Usual stochastic order

Suppose E is partially ordered, with order relation denoted by $\leq$. A function $f : E \rightarrow \mathbb{R}$ is *increasing* if

$$x \leq y \implies f(x) \leq f(y).$$

Definition 2 (Usual stochastic order). *For probability measures $\mu, \nu \in \mathcal{P}(E)$, one says that μ is smaller than ν in the usual stochastic order, and writes*

$$\mu \preceq_{\text{st}} \nu,$$

if

$$\int_E f \, d\mu \leq \int_E f \, d\nu \quad \text{for every bounded increasing } f : E \rightarrow \mathbb{R}.$$

Intuitively, $\mu \preceq_{\text{st}} \nu$ means that a random variable with law ν lies to the right of, or above, a random variable with law μ .

Example 1 (The real line). *On $E = \mathbb{R}$ with the usual order,*

$$\mu \preceq_{\text{st}} \nu$$

if and only if their distribution functions satisfy

$$F_{\mu}(t) \geq F_{\nu}(t) \quad \text{for every } t \in \mathbb{R}.$$

Equivalently, in terms of upper tails,

$$\mu([t, \infty)) \leq \nu([t, \infty)) \quad \text{for every } t \in \mathbb{R}.$$

The measure ν has more mass to the right than μ .

3.2 Convex order

Another important order compares variability rather than location.

Definition 3 (Convex order). *Let $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$, meaning that both measures have finite first moment. One says that μ is smaller than ν in convex order, and writes*

$$\mu \preceq_{\text{cx}} \nu,$$

if

$$\int_{\mathbb{R}^d} \varphi \, d\mu \leq \int_{\mathbb{R}^d} \varphi \, d\nu$$

for every convex function $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$ for which the two integrals are well-defined.

Since affine functions and their negatives are convex, convex order implies that μ and ν have the same barycentre:

$$\int x \mu(\mathrm{d}x) = \int y \nu(\mathrm{d}y).$$

The interpretation is that ν is a mean-preserving spread of μ .

Example 2. On $\mathbb{R}$,

$$\delta_0 \preceq_{\text{cx}} \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1.$$

Indeed, Jensen's inequality gives

$$\varphi(0) \leq \frac{1}{2}\varphi(-1) + \frac{1}{2}\varphi(1)$$

for every convex φ . The second law is more variable, but the mean is still zero.

4 Strassen's theorem for the usual stochastic order

Strassen's theorem says that the integral comparison defining the usual stochastic order is equivalent to the existence of an ordered coupling.

Theorem 1 (Strassen theorem: usual stochastic order). *Let E be a Polish space equipped with a closed partial order $\leq$. In particular, one may take $E = \mathbb{R}^d$ with the coordinatewise order. For $\mu, \nu \in \mathcal{P}(E)$, the following are equivalent:*

(i) $\mu \preceq_{\text{st}} \nu$, i.e.

$$\int f \, \mathrm{d}\mu \leq \int f \, \mathrm{d}\nu$$

for every bounded increasing function f .

(ii) There exists a coupling $\pi \in \Pi(\mu, \nu)$ such that

$$\pi(\{(x, y) : x \leq y\}) = 1.$$

Equivalently, there exist random variables $X \sim \mu$ and $Y \sim \nu$ such that

$$X \leq Y \quad \text{almost surely.}$$

The implication (ii) $\Rightarrow$ (i) is immediate. If $X \leq Y$ almost surely and f is increasing, then

$$f(X) \leq f(Y) \quad \text{almost surely,}$$

and therefore

$$\int f \, \mathrm{d}\mu = \mathbb{E}[f(X)] \leq \mathbb{E}[f(Y)] = \int f \, \mathrm{d}\nu.$$

The nontrivial part is (i) $\Rightarrow$ (ii). It says that all scalar inequalities against increasing test functions can be realised by a single joint construction in which the lower random variable is below the upper one almost surely.

Example 3 (Quantile coupling on $\mathbb{R}$). Let F_μ^{-1} and F_ν^{-1} be the quantile functions

$$F_\mu^{-1}(u) := \inf\{x \in \mathbb{R} : F_\mu(x) \geq u\}, \quad 0 < u < 1.$$

If $\mu \preceq_{\text{st}} \nu$, then

$$F_\mu^{-1}(u) \leq F_\nu^{-1}(u) \quad \text{for every } u \in (0, 1).$$

Taking $U \sim \text{Unif}(0, 1)$ and setting

$$X = F_\mu^{-1}(U), \quad Y = F_\nu^{-1}(U),$$

we get

$$X \sim \mu, \quad Y \sim \nu, \quad X \leq Y \quad \text{almost surely.}$$

This is the monotone, or quantile, coupling.

5 Strassen's theorem for convex order

There is a second fundamental form of Strassen's theorem, where the pathwise constraint $X \leq Y$ is replaced by a conditional expectation constraint.

Definition 4 (Martingale coupling). *Let $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$. A coupling $\pi \in \Pi(\mu, \nu)$ is a martingale coupling if, after disintegrating*

$$\pi(\mathrm{d}x, \mathrm{d}y) = \mu(\mathrm{d}x) \pi_x(\mathrm{d}y),$$

one has

$$\int_{\mathbb{R}^d} y \pi_x(\mathrm{d}y) = x \quad \text{for } \mu\text{-a.e. } x.$$

Equivalently, if $(X, Y) \sim \pi$, then

$$\mathbb{E}[Y \mid X] = X.$$

Theorem 2 (Strassen theorem: convex order). *Let $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$. Then*

$$\mu \preceq_{\text{cx}} \nu$$

if and only if there exists a martingale coupling

$$\pi \in \Pi(\mu, \nu)$$

between μ and ν . Equivalently, there are random variables $X \sim \mu$ and $Y \sim \nu$ such that

$$\mathbb{E}[Y \mid X] = X.$$

Again, one implication follows from Jensen's inequality. If $\mathbb{E}[Y \mid X] = X$ and φ is convex, then

$$\varphi(X) = \varphi(\mathbb{E}[Y \mid X]) \leq \mathbb{E}[\varphi(Y) \mid X].$$

Taking expectations gives

$$\int \varphi \mathrm{d}\mu \leq \int \varphi \mathrm{d}\nu.$$

The converse is the martingale form of Strassen's theorem: the family of convex integral inequalities is exactly the condition needed for the existence of a martingale transport plan.

6 Connection with optimal transport

The link with optimal transport is simple but important: a coupling is a transport plan. Stochastic orders ask whether there exists a transport plan with special structure.

6.1 Order-constrained transport

Let

$$R := \{(x, y) \in E \times E : x \leq y\}$$

be the order relation. Define

$$\Pi_R(\mu, \nu) := \{\pi \in \Pi(\mu, \nu) : \pi(R) = 1\}.$$

Then Strassen's theorem says

$$\Pi_R(\mu, \nu) \neq \emptyset \quad \Longleftrightarrow \quad \mu \preceq_{\text{st}} \nu.$$

Thus the usual stochastic order is precisely the feasibility condition for a transport plan supported on the order relation.

This can be expressed as an optimal transport problem by using the cost

$$c_R(x, y) := \begin{cases} 0, & x \leq y, \\ +\infty, & x \not\leq y. \end{cases}$$

Then

$$\inf_{\pi \in \Pi(\mu, \nu)} \int c_R(x, y) \pi(\mathrm{d}x, \mathrm{d}y)$$

is finite if and only if there exists a coupling supported on R , which is exactly the conclusion of Strassen's theorem.

More generally, once feasibility is known, one may optimise a cost among only ordered couplings:

$$\inf_{\pi \in \Pi_R(\mu, \nu)} \int c(x, y) \pi(\mathrm{d}x, \mathrm{d}y).$$

This is an order-constrained optimal transport problem.

6.2 Kantorovich duality viewpoint

The transport formulation also explains why test functions appear. For a cost c , the Kantorovich dual problem has the form

$$\sup_{\phi, \psi} \left\{ \int \phi \, \mathrm{d}\mu + \int \psi \, \mathrm{d}\nu : \phi(x) + \psi(y) \leq c(x, y) \right\}.$$

For the constraint cost c_R , the inequality

$$\phi(x) + \psi(y) \leq c_R(x, y)$$

only imposes a restriction when $x \leq y$. Taking

$$\phi = f, \quad \psi = -f,$$

the dual constraint becomes

$$f(x) - f(y) \leq 0 \quad \text{whenever } x \leq y,$$

which means precisely that f is increasing. The corresponding dual value is

$$\int f \, \mathrm{d}\mu - \int f \, \mathrm{d}\nu.$$

Thus increasing test functions are the dual certificates for the existence, or non-existence, of an ordered coupling. This is the optimal-transport shadow of Strassen's theorem.

6.3 Martingale optimal transport

For convex order, the constrained set of transport plans is

$$\Pi_M(\mu, \nu) := \left\{ \pi \in \Pi(\mu, \nu) : \int y \pi_x(\mathrm{d}y) = x \text{ for } \mu\text{-a.e. } x \right\}.$$

Strassen's theorem says

$$\Pi_M(\mu, \nu) \neq \emptyset \quad \Longleftrightarrow \quad \mu \preceq_{\text{cx}} \nu.$$

Therefore convex order is the feasibility condition for *martingale optimal transport*. Given a cost c , the martingale optimal transport problem is

$$\inf_{\pi \in \Pi_M(\mu, \nu)} \int c(x, y) \pi(\mathrm{d}x, \mathrm{d}y),$$

or similarly the corresponding supremum problem.

The martingale constraint changes the dual problem. A typical superhedging dual for the supremum problem is

$$\sup_{\pi \in \Pi_M(\mu, \nu)} \int c(x, y) \pi(\mathrm{d}x, \mathrm{d}y) = \inf_{\varphi, \psi, h} \left\{ \int \varphi \, \mathrm{d}\mu + \int \psi \, \mathrm{d}\nu \right\},$$

where the infimum is over functions satisfying

$$\varphi(x) + \psi(y) + h(x) \cdot (y - x) \geq c(x, y) \quad \text{for all } x, y.$$

The additional term $h(x) \cdot (y - x)$ is invisible under martingale couplings, because

$$\int h(x) \cdot (y - x) \pi(\mathrm{d}x, \mathrm{d}y) = 0$$

whenever $\mathbb{E}[Y \mid X] = X$. This is the same mechanism by which affine functions do not distinguish two measures in convex order once their means are equal.

7 Comparison of the main orders

Order	Test functions	Coupling condition	OT interpretation
Usual stochastic order	increasing f	$X \leq Y$ a.s.	support-constrained OT
Convex order	convex φ	$\mathbb{E}[Y \mid X] = X$	martingale OT
Increasing convex order	increasing convex φ	submartingale type constraint	submartingale OT

The common pattern is the following:

$$\text{integral inequalities against test functions} \iff \text{existence of a constrained coupling.}$$

Optimal transport provides the language and tools for studying such couplings. Classical OT optimises over all couplings; Strassen-type theorems identify when the admissible class remains nonempty after imposing order, martingale, or similar structural constraints.

8 Finite-state intuition

On a finite partially ordered set E , the usual stochastic order has a simple network-flow interpretation. A coupling π is a nonnegative matrix with row sums μ and column sums ν . The ordered-coupling constraint says

$$\pi(x, y) = 0 \quad \text{unless } x \leq y.$$

Thus Strassen's theorem becomes a feasible-flow theorem: mass initially located at x may only be transported upward to states $y \geq x$. The condition $\mu \preceq_{\text{st}} \nu$ says exactly that every increasing test function has no larger expectation under μ than under ν , or equivalently that the target law ν has enough mass in every upper region to receive the mass that must be moved upward.

9 Summary

Stochastic orders compare probability measures by testing them against a class of functions. Strassen's theorem converts these analytic comparisons into the existence of couplings with structural constraints:

$$\mu \preceq_{\text{st}} \nu \iff \exists(X, Y) : X \sim \mu, Y \sim \nu, X \leq Y \text{ a.s.},$$

and

$$\mu \preceq_{\text{cx}} \nu \iff \exists(X, Y) : X \sim \mu, Y \sim \nu, \mathbb{E}[Y | X] = X.$$

Since a coupling is the same thing as a transport plan, these results are naturally part of optimal transport. The usual stochastic order corresponds to transport supported on an order relation, while convex order corresponds to martingale transport. Kantorovich duality explains why the relevant test functions appear as dual certificates for the feasibility of these constrained transport problems.

References

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